# Bayesian Inference
## Practice Problems

4. Suppose we need to sample from the bivariate distribution with p.d.f.

$$\pi(x, y) \propto e^{-xy} \mathbb{1}(x, y \in (0, c)),$$

where $c > 0$ is known, and $(0, c)$ denotes the (open) interval between 0 and $c$. Find the full conditional distributions of $x$ and $y$, and write the Gibbs sampling algorithm.

5. Let $X_1, X_2, \ldots, X_n | \mu_0, \mu_1, p \overset{iid}{\sim} F(\mu_0, \mu_1, p)$, where $F(\mu_0, \mu_1, p)$ is the distribution with p.d.f.

$$f(x|\mu_0, \mu_1, p) = (1 - p)\mathcal{N}(x|\mu_0, \lambda^{-1}) + p\mathcal{N}(x|\mu_1, \lambda^{-1}),$$

the usual two-component Normal mixture model. Here, $\mu_0, \mu_1 \in \mathbb{R}$ and $p \in (0, 1)$ are unknown parameters, and $\lambda$ is fixed and known.

Assume the following prior distributions

$$\mu_0, \mu_1 \overset{iid}{\sim} \mathcal{N}(m_0, \ell_0^{-1}) \quad \text{and} \quad p \sim \texttt{Bernoulli}(a_0, b_0),$$

where $m_0, \ell_0, a_0, b_0$ are known hyperparameters. How do you generate the unknown parameters using Gibbs sampling?